

Google Dorking

Estimated time needed: 5 minutes

Learning objectives

- Define Google Dorking
- Explain the use of Google Dorking in pen testing to assess an organization's security posture
- Review basic Google Dork commands, search operators, wildcards, and special characters to improve the accuracy of your online searches

What is Google Dorking?

Google Dorking, or Google Hacking, is an advanced search technique that uses specialized queries to uncover information that is not readily accessible through a standard Google search. By leveraging Google's search capabilities, you can locate hidden data, access confidential documents, and find specific file types online. You can also use these queries to reveal website vulnerabilities, such as exposed databases, login portals, and administrator interfaces.

Ethical use of Google Dorking

Google Dorking can be a powerful tool, but it is essential to use it responsibly and ethically. Using Google Dorks for legitimate purposes, like performing security assessments with proper authorization, is entirely lawful. However, engaging in specific actions, like circumventing paywalls or gaining unauthorized access to restricted pages, can be illegal and amount to intellectual property theft.

Hence, the ethical use of Google Dorks involves:

- Maintaining good intentions
- Avoiding unauthorized access
- Adhering to Google's Terms of Service to prevent legal complications

Google Dorking in pen testing

In penetration testing (pen testing), security professionals primarily use Google Dorking to assess an organization's security posture. Pen testers craft specific search strings to identify potential entry points that cyber attackers could exploit. For example, they might use a query to find a website with a misconfigured server that exposes sensitive directories or files. On discovering these vulnerabilities, organizations can take corrective measures to fortify their security defenses and protect against unauthorized access.

Google Dork commands

Google Dork commands are advanced search queries used to uncover specific information or vulnerabilities. They help identify exposed data or misconfigured website settings. Mastering these commands enhances the precision and effectiveness of your searches.

Here are a few basic commands to get you started.

Google Dork Basic commands

Command	Description	Syntax	Use case
After	Finds content indexed after a specific date	after:[date]	• Find articles published after January 1, 2022 • Example: after:2022-01-01
Allinurl	Finds pages with multiple keywords in the URL	allinurl:[keywords]	• Locate pages with "Python programming" in the URL • Example: allinurl:python programming
Allintitle	Finds pages with multiple keywords in the title	allintitle:[keywords]	• Search for blog posts with "machine learning tutorial" in the title • Example: allintitle:machine learning tutorial
AROUND(X)	Finds pages where two terms are within a specified number of words from each other	[term1] AROUND(X) [term2]	• Find articles where "artificial intelligence" appears close to "ethics" • Example: artificial intelligence AROUND(10) ethics
Before	Finds content indexed before a specific date	before:[date]	• Discovering web pages published before June 15, 2020 • Example: before:2020-06-15
Define	Displays the definition of a word or phrase	define:[word or phrase]	• Find the definition of "quantum computing" • Example: define:quantum computing
Ext	Finds a specific file extension	ext:[file extension]	• Search for files with the .doc extension • Example: ext:doc
Filetype	Locates specific file types like PDF or XLS	filetype:[file type]	• Find PDF documents related to the "annual report" • Example: filetype:pdf annual report
Inanchor	Searches for keywords within the anchor text of links on a web page	inanchor:[keyword]	• Find links with "buy now" in the anchor text • Example: inanchor:buy now

Command	Description	Syntax	Use case
Info	Provides details about a website, including cache and similar pages	info:[URL]	- Obtain detailed information about example.com - Example: info:example.com
Intext	Searches for keywords within the body text of a web page	intext:[keyword]	- Locate web pages with "data science" in the body text - Example: intext:data science
Intitle	Finds a keyword within a web page's title	intitle:[keyword]	- Search for pages with "blockchain" in the title - Example: intitle:blockchain
Inurl	Searches for a keyword within a URL	inurl:[keyword]	- Find URLs that contain "login" - Example: inurl:login
Link	Finds web pages linking to a specific URL	link:[URL]	- Discover pages that link to wikipedia.org - Example: link:wikipedia.org
Map	Shows the map of a location or address	map:[location or address]	- Find the map for "1600 Amphitheatre Parkway, Mountain View, CA" - Example: map:1600 Amphitheatre Parkway, Mountain View, CA
Numrange	Searches for numbers within a specified range	[number]..[number]	- Find products priced between \$50 and \$100 - Example: 50..100
Phonebook	Searches for phone numbers and contact information for a person or business	phonebook:[name or business]	- Look up contact information for "John Doe" - Example: phonebook:John Doe
Related	Displays pages related to a specific URL	related:[URL]	- Identify pages related to nytimes.com - Example: related:nytimes.com
Site	Finds results on a specific website or domain	site:[domain]	- Search for "climate change" articles on bbc.com - Example: site:bbc.com climate change

A thorough understanding of these Google Dork commands and their nuances can help significantly improve the accuracy and efficiency of your online searches.

Combining search operators

Logical operators like AND and OR are essential when constructing effective Google Dork queries. They enable the organization of queries based on logic principles, helping you achieve more accurate and relevant search results.

Search operator	Description	Syntax	Use case
AND	- Combines two or more conditions - All conditions must be true	condition1 AND condition2	- Filter records showing all employees in the sales department over the age of 30 - Example: SELECT * FROM employees WHERE age > 30 AND department = 'Sales';
OR	- Combines two or more conditions - At least one condition must be true	condition1 OR condition2	- Filter records to show all employees who are over the age of 30 or in the sales department - Example: SELECT * FROM employees WHERE age > 30 OR department = 'Sales';

Wildcards and special characters

Wildcards and special characters enhance Google Dork searches by adding versatility through adaptive search techniques. You can use the asterisk (*) and hyphen (-) to broaden or narrow the scope of your queries and achieve more accurate and relevant search results.

Special characters	Description	Syntax	Use case
Asterisk (*)	Wildcard character that can replace any word or phrase in a search query	search_term * search_term	- Find variations of phrases or sentences - Example: site:example.com filetype:pdf security * analysis
Hyphen (-)	Special character used to exclude specific terms from the search results	search_term - excluded_term	- Filter out unwanted results that contain specific terms - Example: site:example.com developer guide - outdated

Summary

In this reading, you learned how Google Dorking can be used to uncover specific information or vulnerabilities.

- Google Dorking is an advanced search technique that uses specialized queries to uncover information that is not readily accessible through a standard Google search.

- Google Dorks can be a powerful tool, but it is essential to use them responsibly and ethically.
- In pen testing, security professionals primarily use Google Dorking to assess an organization's security posture.
- Google Dork commands are advanced search queries that uncover specific information or vulnerabilities.
- Logical operators like AND and OR organize queries based on logic principles, helping you achieve more accurate and relevant search results.
- Wildcards and special characters like the asterisk (*) and hyphen (-) enhance Google Dork searches by adding versatility through adaptive search techniques.

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